

GOKUL KRISHNAN G

+91 9360234599 | gokulkr3331y@gmail.com | Portfolio: [myportfolio-delta-one-21.vercel.app](#)

[linkedin.com/in/gokul-krishnan-g-2026-ai](#) | [github.com/GOKULKRISHNAN2044](#) | Chennai, India (Open to Relocate / Remote)

PROFESSIONAL SUMMARY

AI-Native Software Engineer with **1 year cumulative engineering experience** (including **8+ months production AI** at AVASOFT) and a **B.Tech in AI & Data Science (8.33 CGPA)**. Proven track record in architecting **Autonomous Multi-Agent Systems (LangGraph, FastMCP)**, **Vision LLM Pipelines (AWS Bedrock)**, and **Concurrent Microservices (Go 1.22+, FastAPI, pgvector)** with strict 4-tier Clean Architecture.

TECHNICAL SKILLS

Languages: Python, Go (Golang 1.22+), Java, JavaScript (ES6+), SQL, HTML5, CSS3
AI & Agentic Systems: LangGraph, LangChain, FastMCP (Model Context Protocol), MultiServerMCPClient, RAG, HITL
LLMs & Cloud AI: AWS Bedrock (Claude Haiku 4.5 Vision, Titan Embeddings), OpenAI APIs, Ollama, SBERT
Computer Vision & ML: OpenCV, TensorFlow, Keras CNN, Scikit-Learn, OCR Text Extraction, Image Preprocessing
Backend & Architecture: Clean/Layered Architecture, Gorilla Mux, FastAPI, GORM ORM, SQLAlchemy 2.0, REST APIs
Databases & Vector Search: PostgreSQL, pgvector (Cosine Similarity), MySQL, SQLite, Asyncpg, AsyncPostgresSaver
Cloud & DevOps: AWS (ECS, EventBridge, SES, S3, RDS, CloudWatch, VPC), Docker, Azure DevOps CI/CD, Git, Linux

PROFESSIONAL EXPERIENCE

AVASOFT **Dec 2025 – Aug 2026**
Chennai, India
AI Analyst / Software Engineer (Full-Time — Joined Mid Final Year)
Project 1: AI-Powered Ticketing & Multi-Modal Automation Platform *Live Sports & Music Enterprise Client*

- Engineered event-driven microservices (**FastAPI & Go, 50+ REST APIs**) processing **30,000+ monthly emails**; slashed incident response times from **4+ hours to <45 seconds** via AWS EventBridge, SES, and ECS.
- Built a **multi-modal AI vision pipeline** using **Claude Haiku 4.5 (AWS Bedrock)** and **OpenCV** OCR to parse emails and attachments with **>97% entity extraction accuracy**.
- Configured **Azure DevOps CI/CD pipelines** with automated builds, tests, and container deployments on PR events.

Project 2: T3 – AI Assessment & Developer Evaluation Platform *AVASOFT Internal Core Product*

- Voice AI Interviews:** Built an agentic pipeline with real-time speech transcription and LLM-based rubric candidate scoring.
- MCQ & Live Coding Engine:** Automated question generation; built AST-driven syntax and logic validation via FastAPI.

ENTERPRISE TECHNICAL PROJECTS — Core Systems & Agent Architectures

Enterprise Room Booking Management Microservice (Go Flagship) | *Go 1.22+ · Clean Architecture · GORM · Uber Zap · Docker · Swagger*  GitHub

- Engineered a high-concurrency booking microservice in **Go 1.22+** implementing strict **4-Layer Clean Architecture** (Handlers, Services, Repositories, Entities), GORM transactional persistence, Uber Zap logging, Docker, and Swagger docs.

FastAPI Clean Architecture Production REST API | *Python 3.11+ · FastAPI · SQLAlchemy 2.0 · PostgreSQL · Pytest · UV*  GitHub

- Architected a modular production REST API following strict **Clean Layered Architecture** principles (Routers → Services → Repositories → Models) with Pydantic v2 schemas, async PostgreSQL, and automated Pytest/UV test suites.

Autonomous Multi-Agent Code Review System & FastMCP Framework | *LangGraph · AWS Bedrock · FastMCP · AST Analysis · Python*  LangGraph

- Orchestrated stateful collaborative agents in **LangGraph** with cyclic self-healing AST linting, AsyncPostgresSaver persistent memory, and **FastMCP Human-in-the-Loop** decision checkpoints (place_order) via MultiServerMCPClient.

Enterprise RAG Knowledge Base & Semantic Recommendation System | *FastAPI · AWS Bedrock · pgvector · Cosine Search · Python*  GitHub

- Architected a 2-stage asynchronous RAG pipeline: automated document ingestion, semantic chunking, high-dimensional vector embeddings, and Top-K cosine distance retrieval over **PostgreSQL pgvector** with hallucination-free LLM synthesis.

INTERNSHIPS

Edu-Varsity

Cloud Computing Intern

Role Definition: Architected and deployed scalable, isolated multi-tier cloud infrastructure on AWS for document processing with automated health telemetry. **Project:** Cloud-Native Resume Enhancer Web Platform

- Deployed multi-tier containers on **AWS ECS** inside a custom **VPC** with isolated public/private subnets and **RDS PostgreSQL**.
- Engineered S3 pre-signed upload pipelines, CloudWatch health alarms, and multi-stage **Docker** container image optimization.

Jun 2024 – Jul 2024

Chennai, India

Tech: AWS (VPC, ECS, RDS, CloudWatch, S3, IAM), Docker

FLASH INFOLABS

Full Stack Developer Intern

Role Definition: Engineered full-stack REST API microservices, database schemas, and interactive user directories for corporate workforce operations. **Project:** Enterprise Employee Management & Directory Portal

- Built modular backend REST APIs in **Python FastAPI** with **Pydantic v2** schema validations and relational database queries.
- Implemented **RBAC access security**, standardized JSON responses, and responsive single-page search interfaces via modern JavaScript.

Oct 2023 – Nov 2023

Chennai, India

Tech: Python FastAPI, PostgreSQL, JavaScript, HTML5, CSS3, REST APIs

ACADEMIC & RESEARCH CAPSTONES — University Engineering Foundations

AI-Powered Psychometric Assistant for Stress | Python · SBERT · DistilBERT · Gemini API · Rasa

- Developed an adaptive stress evaluation assistant using **SBERT** sentence embeddings, **DistilBERT** classification, and **Gemini API** tone analysis integrated with an interactive Rasa conversational chatbot.
- Engineered a real-time stress assessment dashboard featuring speedometer visualizations and question sequencing.

Automated Guided Vehicle (AGV) — Intelligent Navigation | Python · OpenCV · Path Planning · NumPy

- Engineered a computer vision trajectory planning and autonomous navigation system for automated guided vehicles using **OpenCV** image processing and **NumPy** path planning algorithms.
- Implemented real-time obstacle avoidance and edge tracking; open-source codebase **forked by 2 external community contributors**.

Presence Pro — Face Recognition & Biometric Attendance | Python · OpenCV · Haar Cascades · SQLite

- Implemented real-time biometric face verification using **OpenCV Haar Cascades** and 128-d face encodings linked to SQLite.
- Automated absentee notifications via **SMTP** email protocols with timestamped attendance audit logs and summary reports.

ACHIEVEMENTS & RESEARCH PUBLICATIONS

🏆 1st Prize — Ideathon | RESONANCE '25, Rajalakshmi Engineering College (IEEE Computer Society Chapter)

Feb 2025

📄 2 International Research Publications | IJFMR (IF: 9.24, 2025) & TIJER (IF: 8.57, Jul 2024)

2024 – 2025

🏠 IEEEExtreme 18.0 | Global programming competition, ranked among 19,000+ participants

Oct 2024

EDUCATION

Panimalar Engineering College

B.Tech in Artificial Intelligence & Data Science (CGPA: 8.33 / 10)

Jayam Metric Higher Secondary School

Higher Secondary Certificate — Computer Science (88.34%)

2022 – 2026

Chennai, India

May 2022

Dharmapuri, India

KEY CERTIFICATIONS

☀️ Oracle Cloud Infrastructure 2024 Generative AI Certified Professional | Oracle University

Jul 2024 – Jul 2026

☀️ NPTEL — Introduction to Internet of Things | IIT Kharagpur / SWAYAM (Elite Certificate)

2024

☀️ Machine Learning Training | AcmeGrade × IIT Bombay Mood Indigo

2023